Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/22/2024
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Project Title:	Diffusion Coefficients of Dopants in Si and Ge Melts		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60616-3729	**Congressional District:**	7
Comments:	Principal Investigator was previously at Rensselaer Polytechnic Institute.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	08/07/2020	**End Date:**	08/06/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Volz, Martin Ph.D. (NASA Marshall Space Flight Center)		
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Performance Goal Text:			
	The compositional homogeneity and electronic properties of single crystal semiconductors depend on the diffusion coefficients of impurities and dopants dissolved in the melt. However, even after several decades of research, the published values of diffusion coefficients can vary significantly. Diffusion is a slow process, and thus affected by minute levels of convection. Thus, it is expected that previous measurements of diffusion have been compromised by natural or buoyancy-driven convection, which is unavoidable on Earth. Microgravity presents an optimal environment to obtain high-fidelity diffusion data. This data is essential for the computer models which are used to optimize industrial crystal growth processes. The key goals of this proposal are as follows: (i) Develop and calibrate a process to measure diffusion coefficients on Earth.		

Task Description:	(ii) Use this process to make accurate measurements of diffusion coefficients in Ge in microgravity. (iii) Use this process to measure the diffusion coefficients of impurities in Si melts. We propose to conduct on Earth and in microgravity the following experiments: (a) Bridgman growth experiments using bundles of small diameter ampules. (b) Direct measurements of diffusion coefficients in bundles of capillary tubes. The key goal is to determine precisely the diffusion coefficients of Ga, B, Sb, and Si in germanium. Furthermore, on Earth, following the pioneering Burton-Prim-Slichter (BPS) papers [1,2] we propose to conduct (c) Czochralski crystal growth experiments, at high RPM, from small melts, in order to obtain steady laminar symmetric flow (eliminate unsteady natural convection) and use the BPS model and finite element (FE) modeling, to measure the diffusion confidents in Ge and Si melts. In preparation for the microgravity experiments, the following ground based experiments will be conducted: (i) Direct measurements of diffusion coefficients in spiral or circular capillary tubes in the 5 Tesla axial magnet at Marshall Space Flight Center (MSFC). (ii) Direct measurements of diffusion coefficients using bundles of straight capillary tubes in 1 Tesla transverse magnet at Illinois Institute of Technology (IIT). (iii) Bridgman growth experiments using a bundles of small diameter ampoules inside the Sample Cartridge Assembly (SCA). The microgravity experiments will utilize the Low Gradient Furnace (LGF) furnace in the Materials Science Research Rack (MSRR). The experiments will consist of both “direct” measurements of diffusion using small capillaries as well as directional solidification experiments. The variety of experiment variables will provide a surfeit of relevant diffusion data in alignment with the goals of the MaterialsLab initiative. These include the variety of dopants to be measured, with varying segregation coefficients, the variation in processing parameters such as temperature, and the ability to process up to 7 small capillaries in a single experiment. References: [1] J.A. Burton, R.C. Prim, W.P. Slichter. The Distribution of Solute in Crystals Grown from the Melt. Part I. Theoretical. Journal of Chemical Physics 21:1987, 1953. [2] J.A. Burton, R.C. Prim, W.P. Slichter. The Distribution of Solute in Crystals Grown from the Melt. Part II. Experimental. The Journal of Chemical Physics 21:1991, 1953.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We considered and tested several design options for the ampoule with the capillaries, which are suitable for both ground-based measurements of diffusion coefficients, and the flight experiments. Our present focus is on using relatively short “finite” capillaries (typically 40 mm long), which appear to be superior to the longer “semi-infinite” capillaries. The short capillaries are particularly suitable for experiments on the International Space Station (ISS). The ground-based experiments with molten Si require specialized equipment because of the high melting point of Si (T_m = 1420 °C). The experiments are conducted on the commercial Arthur D. Little (ADL) Czochralski puller (as described in the original 2016 proposal). Furthermore, in 2024, we will have designed and built a tabletop high vacuum tubular capable of melting Si. The heaters in this furnace are in vacuum, allowing the use of a tantalum wire that melts at 3017 °C.
Bibliography Type:	Description: (Last Updated: 05/30/2024)
Abstracts for Journals and Proceedings	Ostrogorsky AG, Volz M, Croell A. "Measurement of diffusion coefficients of dopants in Ge and Si melts." Materials Research Society Fall Meeting, San Diego, California, February 26, 2020. Abstracts. Materials Research Society Fall Meeting, San Diego, California, February 26, 2020. , Feb-2020
NASA Technical Documents	Ostrogorsky AG, Volz M, Croell A. "Measurement of diffusion coefficients of dopants in Ge and Si melts." Huntsville, Alabama: NASA Marshall Space Flight Center, 2020. NASA/No. M20-7971. https://ntrs.nasa.gov/api/citations/20200001746/downloads/20200001746.pdf , Feb-2020